

Project Design Phase
Solution Architecture

Date	09/03/2026
Team ID	SWTID-2026-4275
Project Name	SpendWise
Maximum Marks	4 Marks

Solution Architecture:

The SpendWise application uses a simple and modular backend architecture to help beginners learn REST API development, JWT authentication, and MongoDB integration. It is built with Node.js, Express, and MongoDB following a clean project structure.

The system follows a client-server architecture, where tools like Postman or Thunder Client send HTTP requests to the backend API. The backend processes authentication, handles business logic, and interacts with MongoDB to store and manage user data.

Goals of the Solution Architecture:

Identify the Best Tech Solution:

- Use **Node.js** and **Express.js** to develop REST APIs.
- Use **MongoDB** with **Mongoose** for database management.
- Implement **JWT** for secure user authentication.
- Test APIs using **Postman** or **Thunder Client**.

Define Structure & Characteristics:

- Use a modular structure with folders for **controllers, models, routes, and middleware**.
- Ensure the system is **simple, maintainable, and scalable**.

Outline Features & Development Phases:

- Project setup and database connection.
- Implement **user registration and login with JWT**.
- Add **JWT middleware** for protected routes.
- Implement **CRUD operations** for user data.
- Test APIs and prepare basic documentation.

Establish Specifications for Development & Delivery

- The backend will provide **REST APIs** for authentication and **CRUD** operations.

